



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Fourth Semester- Diploma in Engineering (Electrical Engineering) Session 2020-21

SPECIALIZATION – Electrical Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Mid Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	ME-4001	GENERAL MECHANICAL ENGG	4	1	2	6	70	20	10	30	20	150
2	EE-4001	ELECTRICAL MACHINES – II	4	1	2	6	70	20	10	30	20	150
3	EE-4002	GENERATION, TRANSMISSION AND DISTRIBUTION	4	1	2	6	70	20	10	30	20	150
4	EE-4003	ELECTRICAL ENGG. DRAWING	4	1	2	6	70	20	10	30	20	150
5	DE-4005	ENTREPRENEURSHIP / MARKETING MANAGEMENT	4	-	-	4	70	20	10	-	-	100
6	EE-4004	PROFESSIONAL ACTIVITIES -	-	-	2	-	-	-	-	-	-	-
		Total	20	04	08	28	350	100	50	120	80	700

Theory Credits: 24
Practical Credits: 04
Total Credits: 28

Theory Marks: 350
Practical Marks: 200
Quiz, Mid Term, : 150

Total: 700



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

DIPLOMA IN ELECTRICAL ENGINEERING

(Courses effective from Academic Year 2019-20)

SYLLABUS OF COURSES TO BE OFFERED “DIPLOMA IN ENGINEERING”

Disclaimer: The postgraduate programme in Diploma in Engineering syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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**DIPLOMA IN ELECTRICAL
ENGINEERING
Branch - Electrical Engineering**

ME-4001 GENERAL MECHANICAL ENGG

UNIT-I

ENGINEERING MATERIALS : Introduction of Materials , Need and classification of engineering materials , Metals and alloys ,Ferrous Metals (i) Cast Iron (ii) Wrought Iron (iii) Steel (iv) Alloy Steel 1.3.2 Non Ferrous Metals (i) Aluminum (ii) Copper (iii) Lead (iv) Tin (v) Copper tin-antimony alloy (vi) Bearing Metals (vii) Copper tin alloy (viii) Zinc (ix) Copper Zinc alloy

MECHANICAL PROPERTIES AND TESTS: Properties of Materials (i) Stiffness (ii) Strength (iii) Ductility (iv) Malleability (v) Elasticity (vi) Plasticity (vii) Toughness (viii) Brittleness (ix) Hardness and Harden ability (x) Fatigue, Material Test (i) Tensile Test (ii) Impact Test (Izod and charpy) (iii) Hardness Test (Brinell, Rockwell and Vickers)

UNIT-II

HEAT TREATMENT : Definition and objectives of Heat treatment, Effect of different factors in heat treatments, Heat treatment Process (i) Annealing (ii) Normalizing (iii) Hardening by Quenching (iv) Tempering (v) Case hardning (vi) Carburizing

THERMODYNAMICS : 4.1 Introduction 4.1.1 Work, Heat & Power 4.2 Various thermodynamics properties 4.3 Thermodynamic system 4.4 Thermodynamic (i) State of the System (ii) Process on the system 4.5 Statement of Ist and IInd law of thermodynamics 4.6 Law of Ideal gases. (i) Boyle's Law (ii) Charles Law (iii) Gas equation 4.7 Properties of steam (i) Enthelpy of Dry and wet steam (ii) Specific volume of dray and wet steam (iii) Internal Energy of Dry and Wet Steam, Boilers, Classification of boilers (i) Fire tube (ii) Water tube 4.82 Sketch and description of (i) Simple vertical boiler (ii) Lankashire boiler (iii) Babcock and Wilcox boiler (iv) Locomotive boiler.

UNIT-III

I.C. ENGINE :Define Heat Engine,Differentiate I.C. Engine and E.C. Engine, Classification of I.C. Engines. Explain the working of two strokes and four stroke petrol engine with line diagram Explain the working of two stroke and four stroke diesel engine with line diagram,(i) Indicated Horse Power (1HP) (ii) Brake Horse Power (B HP) (iii) Mechanical Efficiency

AIR COMPRESSOR : Introduction of Air Compressor and their classification,Working principle of reciprocating Air-compressor. Industrial uses of Air-compressor 6.4 Multistage reciprocating compressor & their merit & Demerit 6.5 Rotary compressor .

UNIT-IV

FLUID MECHANICS : Definition of various fluid properties ,Fluid pressure and its measurement , Pascal's Law Static Pressure (i) Intensity of pressure at a point in fluid at rest (ii) Pressure head (iii) Absolute and gauge pressure Simple and differential U type mano meters. Total and center of pressure on the plate surface immersed in water Horizontally and vertically.

HYDRODYNAMICS: Energies in fluid (i) Pressure energy (ii) Kinetic energy (iii) Potential energy (iv) Total energy Bernoullis theorem, its assumption and application ,Pitot tube ,Venturimeter ,Orifice meter ,Working principle of Hydraulic Pumps (i) Reciprocating pump (ii) Centrifugal pump , Working Principles of water turbine (i) Impulse turbine (ii) Reaction turbine.

UNIT-V

POWER TRANSMISSION : Methods of Power transmission Belt drive (i) Open and cross belt drive (ii) Application and advantages of belt drive (iii) Velocity ratio of pulleys (iv) Compound belt drive (v) Effect of slip in the belt drive ,Gear drive (i) Simple gear drive (ii) Compound gear drive (iii) Worm and wheel (iv) Bevel gear Velocity ratio in gear drive Merit and demerits of gear drive ,Simple problems of gear drive.

SYLLABUS- DIPLOMA IN ELECTRICAL ENGINEERING, BHABHA UNIVERSITY

**DIPLOMA IN ELECTRICAL
ENGINEERING
Branch - Electrical Engineering**

LIST OF EXPERIMENTS:-

1. Perform Tensile Test of standard mild steel and C.I. specimen
2. Perform Hardness Test Brinell and Rockwell.
3. Impact Test Izod and Charpy on mild steel specimen.
4. Study of Boilers
 - a. Fire tube b. Water tube c. Babcock & Wilcox Boiler d. Boiler mountings
 - e. Boiler accessories.
5. Study of steam engine
6. Study of simple gear & compound train in power transmission system
7. I.C. Engines a. Study of I.C. Engine Two stroke and four stroke Petrol & Diesel Engine
8. Air Compressor Study of Air Compressor, Single stage and multistage compressor their construction and their uses.
9. Fluid Mechanics & Machines
 - (i) Pressure measurement by manometer
 - (ii) Determination of coefficients of discharge of the following devices –
 - a. Venturi meter b. Rota meter
 - c. Orifice Meter d. Pitot tube
 - (iii) Study of Centrifugal & reciprocating pumps.

References Books:-

1. General Mechanical Engineering by S.B. Mathur
2. Elements of Mechanical Engineering by Mathur, Mehta & Tiwari
3. Elements of Mechanical Engineering by Raw & Choudhary
4. Fluid Mechanics by R.S. Khurmi.

**DIPLOMA IN ELECTRICAL
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EE-4001 ELECTRICAL MACHINES – II

UNIT-I

Three phase Induction Motor - Production of rotating magnetic field, principle, construction and types of induction motors. Equivalent circuit, torque equation, torque-slip characteristics. Types of starters: DOL, Star-delta, Autotransformer type, rotor resistance type, contactor type starter. Speed control. No load and blocked rotor test, losses and efficiency. Braking and applications. Simple numerical.

UNIT-II

Synchronous motor - Principle, construction, phasor diagram, effect of change in excitation, V curves, synchronous condenser, starting of motors, hunting and its prevention, coding of synchronous machines.

Synchronous generator - Principle, construction, salient and cylindrical rotors, speed-frequency relationship, EMF equation, distribution and pitch factor, equivalent circuit, synchronous impedance, regulation, O.C.C. and S.S.C., load characteristics, phasor diagram, parallel operation. Methods of synchronization, power-angle characteristics.

UNIT-III

Single phase induction motors - Principle, double revolving field theory. Types of motors with their construction, characteristics and applications. Comparison of three phase with single phase induction motors.

UNIT-IV

AC commutator motors - Introduction, series motor, compensated series motor, commutating poles, universal motor, repulsion motor.

UNIT-V

Special purpose machines - Induction motor, stepper motor, PM motor.

**DIPLOMA IN ELECTRICAL
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LIST OF EXPERIMENTS:-

1. Study of three phase induction motor (parts).
2. Measurement of slip of three phase induction motor.
3. Study of three phase induction motor starters.
4. Study of synchronous machine (parts).
5. OCC and SCC of synchronous generator and determination of regulation.
6. To plot V curves of synchronous motor.
7. Study of different single phase induction motors (construction).
8. Study of AC commutator motors (construction).
9. Study of special purpose motors (construction)

References Books:-

1. Electrical Technology Vol. II BL Thereja Khanna publisher
2. Electrical Machines Bhattacharya T.T.T.I.
3. Electrical Machines Nagrath & Kothari PHI
4. Electrical Machines Vol. I & II PS Bhimbhra Khanna publishers

**DIPLOMA IN ELECTRICAL
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EE-4002 GENERATION, TRANSMISSION AND DISTRIBUTION

UNIT-I

Non Conventional Sources Of Energy - Concept and need of primary and secondary energy sources, difference between conventional and non-conventional sources of energy, concept of solar , wind, biogas, ocean, tidal, geothermal, fuel cell , MHD and their practical applications.

Conventional Sources Of Energy - Detailed study of generating stations - thermal, hydro, nuclear, schematic diagram, site selection main components and auxiliaries for above power stations. Study of gas turbines plant and diesel power plant. Advantages, disadvantages of thermal hydro, nuclear, gas turbine plant and diesel power plant.

UNIT-II

Concept Of Load - Types of load, load curve, load duration curve, connected load, demand factor, average load, maximum demand, load factor, diversity factor, plant utilization factor, capacity factor, reserve capacity. Simple numerical on above terms. Types of Tariff, flat rate, block rate, two part, maximum demand and power factor tariff. Their merits and demerits. Simple problems on above terms.

UNIT-III

Concept of Transmission, single line diagram of complete power system, standard voltages of A.C. Transmission, efficiency (no derivation). H.V.D.C. transmission system, line diagram, advantages and Disadvantages of H.V.D.C. Sag, causes & effects of sag on transmission line, effect of wind, ice and temperature on sag. Types of line supports, type of joints, looms, earth wires, ground wire and vibration dampers. Importance of R,L,C in transmission line (no derivation), skin effect, transposition, corona, advantages and disadvantages of corona, methods of reducing corona, types of insulators, string efficiency and voltage distribution, grading ring and Arcing horn.

UNIT-IV

Types of Transmission line, T and Π network of medium Transmission line, transmission efficiency, Ferranti effect, simple problems of short and medium Transmission line.

Difference between overhead line and underground cables. Classification and construction of L.T. and H. T. cables, Methods of laying.

UNIT-V

Classification of distribution system, ring main, radial and interconnected system. Concept of feeder, distributor and service mains in distribution system. Simple problems.

**DIPLOMA IN ELECTRICAL
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LIST OF EXPERIMENTS:-

1. Study of solar cooker.
2. Study of solar water heater.
3. Study of solar photo-voltaic cells.
4. Study of wind mill.
5. Study of Bio Gas plant.
6. Study of steam power plant, hydro power plant, nuclear power plant.
7. Study of line supports and insulators.
8. Determination of string efficiency of insulator string. Performance of short/ medium transmissions line.
9. Study of L.T. and H.T. Cables and over head conductors.
10. Voltage distribution in radial and ring main system.
11. Visit to a • Substation. • Generating station. • Places where solar, wind, Biogas and tidal power plant are installed.

References Books:-

1. Non Conventional energy sources By G.D. Rai, Khanna publisher
2. Electrical Power By S.L.Uppal, Khanna publisher
3. Electrical Power By J.B. Gupta
4. Power System By V.K. Mehta

**DIPLOMA IN ELECTRICAL
ENGINEERING**
Branch - Electrical Engineering
EE-4003 ELECTRICAL ENGG. DRAWING

UNIT-I

Symbols and Notations - Symbols of practical units, multiples and submultiples, types of supplies, single phase, three phase three wire, three phase four wire, D.C. supply etc. Accessories like main switches, distribution boards, fans, light fixtures, bell, buzzer, lighting arrestor. All types of motor starters, instruments, electronic components etc. Rating plate of machines.

Domestic Wiring - All types of light circuits: Fluorescent tube circuits, intermediate switch circuits, fan circuits. Wiring of a residential building. Sodium vapor lamp, mercury vapor lamp.

UNIT-II

Domestic Wiring - All types of light circuits: Fluorescent tube circuits, intermediate switch circuits, fan circuits. Wiring of a residential building. Sodium vapor lamp, mercury vapor lamp.

Winding Diagrams - Simplex type lap and wave diagrams for D. C. Machines. Single phase and three phase motor winding diagrams.

UNIT-III

Electrical Machine Drawing - Parts of D.C. machines like, magnetic poles, commutator, armature etc. A.C. machines rotor, slip rings, etc. Various cable sections. Bushing of the transformer. Assembly diagrams of D.C. machine, A.C. machine, and transformer.

Power Wiring - Internal wiring diagrams of single phase motor. wiring diagrams of D.C. and A.C. motor starters like three point shunt motor starter, four point compound motor starter, direct on line (D.O.L.) starter, star- delta starter, contactor type and auto transformer starter. Internal connections of D.C. series, shunt and compound motors. Three phase motors: squirrel cage, slip ring, synchronous etc. Plate earthing and Pipe earthing as per I.S.S.

UNIT-IV

Simple Electronic Circuits - Battery eliminator, battery charger, single stage transistor amplifier, connections of common emitter, collector and base amplifier circuits.

Alternator Panel Diagrams - Panel diagram with circuit breaker, isolator, measuring instruments, synchroscope. Over current and earth fault protection, differential protection, voltage regulator etc.

UNIT-V

Transmission And Distribution - All types of transmission towers and distribution poles. Arrangement of various types of cross arms, with insulators, jumpers. Electrical layout of 33KV/ 11KV substation, 11KV/415V pole mounted substations with all protective devices etc.

References Books:-

1. A text book of Electrical Drawing .by S.L. Uppal Khanna pub.
2. Electrical Drawing by K.L. Narang.
3. Electrical Drawing by C.R. Bargan .

DIPLOMA IN ELECTRICAL ENGINEERING

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DE-4005 ENTREPRENEURSHIP / MARKETING MANAGEMENT

UNIT-I

INTRODUCTION TO ENTREPRENEURSHIP Definition of Entrepreneur / Entrepreneur, Difference between Entrepreneurship / Entrepreneurship, Need for Entrepreneurship, qualities of successful entrepreneur, Myths about Entrepreneurship, Classification of entrepreneurs on the basis of different criteria, Reasons for the failure of entrepreneurs.

INDUSTRIES AND BUSINESS ORGANIZATIONS Concept of Industry or Enterprise • Classification of Industries (a) On the basis of capital investment - Tiny (Micro) Industry - Small Scale - Medium Scale - Large Scale (b) Others - Rural Industry - Cottage Industry (c) Forms of Business Organization - Proprietorship - Board & Co-operative - Partnership - Public Ltd. - Private Ltd. - IT Sector - Government Co-operative / Undertakings (d) Tiny small scale Industry - Definition - Its significance in National Development. - Govt. policies for SSI promotions - Sector / Product for SSI.

UNIT-II

INSTITUTIONAL ASSISTANCE (a) Types of Institutional assistance - Infra - structural assistance - Technical Assistance - Financial assistance - Marketing Assistance (b) Information / guidance & Training - SISI - ASK - MPCON - CSIR - CED- MA - NRDC (c) Infrastructure - D/C - AVN/AKVN (e) Finance - SIDBI - KVIB MPFC - NABARD - MPWDC NSIC M.P.A.V.V.N. (d) Marketing - MP- AGRO - NSIC - PM.LUN - EXPORT COPPORATION - KVIP - MPHSVN MPLDC (e) Quality Control - BIS - FPO - MPLUN F.D.A. - AG. MKT. Board INCENTIVES/CONCESSION / FACITLITIES AVAILABLE • Seed money • Incentive / subsidies • Others (Phones, Lands etc)

UNIT-III

PLANNING OF AN INDUSTRIAL UNIT (SSI) • Pre- Planning Stage - Scanning the environment - Market survey - Seeking information - product / project selection • Implementation Stage - PPR Preparation - DIC registration - Arrangement of Land - Arrangement of Power - Obtaining NOC / Licenses from various departments - DPR Preparation - Seeking financial assistance - Commercial Production • Post Implementation stage - Permanent registration from D.I.C. - Availing Subsidies - Diversification / Modification - Setting up of marketing channel / Distribution.

UNIT-IV

ACHIVEMENT MOTIVATION • Historical perspective • Concept of achievement motivation • Significance of achievement motivation • Development of achievement motivation

UNIT-V

FINANCIAL MANAGEMENT OF AN INDUSTRIAL UNIT (SSI) • Tools of financial analysis • Ratio analysis • Fund Flow / Cash flow analysis • Working capital and concepts • Financial accounting

References Books:-

1. Entrepreneurial Development Vol. I,II,III By Vasant desai Himalaya Publicaton
2. CEDMAP (Center of Entrepreneurial development Madhya Pradesh)
3. Udyamita Vikas By Anand Prakashan